

PRESSURE SENSORS

Smartsac, Netherlands

Absolute / Differential / Gauge type / Output - Analogue / Digital / Bridge

Humidity Sensor

Long Term Stability

- Linear 0-100% RH
- Close tolerance
- High performance
- Low Cost

Hall Sensors

(USA make)

www.magnesens.com

Introducing NEW SENSORS

General-Purpose Hall Effect Latch / Open-Drain pre-driver Typ, 30 Gauss

MH189

(Ultra High Sensitivity Sensor) / Chopper Stabilized amplifier stage / A short circuit protected / Pull-up resistor output / Ultra High sensitivity Typ, 15 Gauss

Pin to Pin match

MH187 to US181 / US2881 / A3282 / A3280

MH189 to EW512 / EW372

MH248ESQ - AH180-SNG-7 / AH180-FJG-7

Ultrasonic sensor module

Range of measurement: 0.03 - 3M

Rated working voltage: 5 VDC

Frequency of sensor: 40 KHz

Ultrasonic Detector

Detect distance range: 150-6000MM

Measuring mode: direct reflection

(liquid level, solid level) 5-18VDC

RF Modules

Taiwan

ASK (434.215MHz) / FSK (434.468MHz) / Transceiver (2.4GHz)

PPTC Fuses -CE, UL approved

16V/30V/60V/90V/250V available

SMD & Radial types

RELAY

-CE, UL, TuV, CQC certified

Telecom / General / Automotive relay

Our brand	Sunhold	Goodsky
HL58-T78 12V/24V/48V	RAS120/120/240/48	RWSH120/120/240/48
HL58-T09 12V	R72 120	GSU 50 230
HL58-T40 12V	TNA 1 C 120L	MSH112L
HL58-T91 12V	RTD 120A	GSU1112D

HL58-T78 15V X 12.5 X 14mm (ISO 20x20x10VDC) Miniature relay HL58-T73 12V (Yellow) same like Leon/YOY brand

Strict quality control, guaranteed performance, within time delivery, competitive price.

Telecom Relay

Other Sensors :

Laser / Reflective / Ultrasonic / Photodiode / IRLD / Phototransistor / Fluid / PIR / Gas

Micro Impex

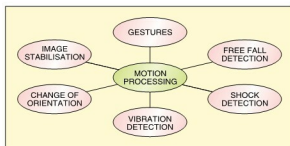
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Types of motion processing

MEMS accelerometer is suitable for a particular application. The integration of more than one type of motion sensing is very common. Most designers have the misconception that a product can only have either linear sensing or rotational sensing. It can be designed by using either an accelerometer or gyroscopes. However, both used in conjunction give better accuracy and reliability. The inertial measurement unit (IMU) that has a multi-axes gyroscope embedded with a multi-axes accelerometer is used in many applications for better accuracy.

Motion sensing for application developers

A motion sensor can be used for multiple applications in the same product. Fig. 1 describes various elements of motion sensing that can be useful for an application in a particular device. Applications are usually software programs that produce a reaction to a sensed motion signal from the accelerometer or gyroscope. Designers opt for high-end MEMS sensors in order to develop multiple applications in the same device. Each of the applications require different data sampling rates and anti-aliasing measures that ensure data accuracy through different filters for a particular application.

Gestures. Gestures normally refer to hand or body movements. Gestures are recognised by the motion sensors and the data is communicated for use in applications. This makes products more usable by eliminating the need for buttons and switches to perform different tasks. Defined gestures such as shakes, taps, double taps, etc, are

used for activating various controls. Button-free designs can be used in products like mobile handsets making it more durable and reducing the cost of the overall system.

Gesture controls are widely used in gaming devices. The 3D mouse

and remote are other applications where gestures are used. Tap recognition is used in wearable medical devices like hearing aids. Shakes are uniquely used in scrolling or 'move to next item' applications in mobile handsets.

Titan is currently working on using gesture control for its new application called the 'active dial'. The normal gesture of checking the time on your wrist watch will make the dial active, else it will remain in an idle state or in switched-off mode, according to B.V. Nagaraj, general manager, Innovedge, Titan Watches. A person's swinging hand movements in the to-and-fro direction is used in the pedometer on the wrist watch to count the steps taken. "Accelerometers as sensors will play an important role in Titan's forthcoming technologies and design," Nagaraj adds. The research group at Titan is working on using motion sensing innovatively in watches.

Free-fall detection. Acceleration is measured along three axes (X, Y, Z

History

MEMS started off as an academic project 20 years ago, and is currently used in a significant number of consumer products. Motion sensing technology was first used in the 1950s in an aircraft navigation system. With continuous research in micro and nanotechnology, MEMS-based motion sensors like accelerometers and gyroscopes have become the key innovation drivers in consumer products. MEMS versions of accelerometers were first used in the 1990s in vehicle airbags as crash sensors. The most recent examples are their use in Nintendo's Wii controller and Apple's iPhone and iPod devices.